Gamification Domain Ontology - Extended Concepts (GaDO-Full)

Domain: Gamification
Date: Sep 15, 2015
Level of formality: Semi-formal.

Conceptualized and Implemented by: Diego Dermeval

Purpose: Present extended concepts related to gamification, gamification best practices, and human needs and behavior involving these concepts.

Scope: Gamification concepts, gamification best practices, game mechanics, psychological needs and human behavior, player models.

Sources of Knowledge:

- Werbach, Kevin, and Dan Hunter. *For the win: How game thinking can revolutionize your business*. Wharton Digital Press, 2012.
- Hamari, Juho, Jonna Koivisto, and Harri Sarsa. "Does gamification work?--a literature review of empirical studies on gamification." *System Sciences (HICSS), 2014 47th Hawaii International Conference on*. IEEE, 2014.
- Seaborn, Katie, and Deborah I. Fels. "Gamification in theory and action: A survey." *International Journal of Human-Computer Studies* 74 (2015): 14-31.
- Mora, Alberto, et al. "A literature review of gamification design frameworks."
- Challco, Geiser Chalco, et al. "Towards an Ontology for Gamifying Collaborative Learning Scenarios." *Intelligent Tutoring Systems*. Springer International Publishing, 2014.

Integration

This ontology makes use of the GaDO Core Ontology.

Important Terms

- Activity Loop
- Engagement Loop
- Progressive Loop
- Best Practice
- Emotions
- Feedback
- Game Design Element
- Metric
- Motivation and Need Theories
- Self-Determination Theory
- Autonomy

- Competence
- Relatedness
- Motivational Affordance
- Narrative
- Player Action
- Target Behavior
- Player Model
- BrainHex
- Motivational Components (Yee)
- Player Type
- Achiever
- Conqueror
- Daredevil
- Mastermind
- Seeker
- Socializer
- Survivor
- Relationships
- Target Behavior Category
- Competition
- Effectiveness
- Enjoyment
- Exploration
- Participation
- Performance

Classes and SubClasses

- ActivityLoop
    - EngagementLoop
    - ProgressiveLoop
- BestPractice
- Emotions
- Feedback
- GameDesignElement
- Metric
- MotivationAndNeedTheories
    - SelfDeterminationTheory
        - SdtAutonomy
        - SdtCompetence
        - SdtRelatedness
- MotivationalAffordance
- Narrative
- PlayerAction
- TargetBehavior

- PlayerModel
    - BrainHex
    - YeeMotivationalComponent
- PlayerType
    - Achiever
    - Conqueror
    - Daredevil
    - Mastermind
    - Seeker
    - Socializer
    - Survivor
- Relationships
- TargetBehaviorCategory
    - Competition
    - Effectiveness
    - Enjoyment
    - Exploration
    - Participation
    - Performance

Properties

- Competition: competitionName
- Effectiveness: efferctivenessName
- Enjoyment: enjoymentName
- Exploration: explorationName
- Participation: participationName
- Performance: performanceName
- ProgressiveLoop: loopDifficult
- EngagementLoop: loopQuality, loopStatus
- Metric: metricDescription, metricExpectedValue, metricValue

Properties' Aspects/Types

- competitionName: string
- efferctivenessName: string
- enjoymentName: string
- explorationName: string
- participationName: string
- performanceName: string
- loopDifficult: float[0.0-1.0]
- loopQuality: float[0.0-1.0]
- loopStatus: boolean
- metricDescription: string
- metricExpectedValue: double
- metricValue: double

Binary-Relations

Table: Binary-Relations

Relation Name	Source Concept	Target Concept	Inverse Relation
coversComponent	PlayerType	GameDesignElement	-
describesType	PlayerModell	PlayerType	-
hasCategory	TargetBehavior	TargetBehaviorCategory	-
hasGameElement	BestPractice	Component, Mechanic	-
hasMetric	TargetBehavior	Metric	isUsedFor
hasPlayer	PlayerAction	Player	-
hasTargetBehavior	BestPractice	TargetBehavior	-
isAccomplishedWith	TargetBehaavior	ActivityLoop	leadsTo
isComposedBy	EngagementLoop	PlayerAction, MotivationalAffordance, FeedBack	isPartOfEngagementLoop
isMotivatedWith	YeeMotivationalComponent	GameDesignElement	-
isPartOfEngagementLoop	FeedBack, MotivationalAffordance, PlayerAction	EngagementLoop	isComposedBy
leadsTo	ActivityLoop	TargetBehavior	isAccomplishedWith
leadsToCompetition	Leaderboards	Competition	-
leadsToEffectiveness	Leaderboards, Badges	Effectiveness	-
leadsToEnjoyment	Theme, Avatars, Rewards, Achievements, Badges, Story, Quests, Points	Enjoyment	-
leadsToExploration	BossFights, Challenges, Levels, Achievements	Exploration	-
leadsToParticipation	Badges, Challenges, Themes, Points, Story, Levels,	Participation	-

	Leaderboards, Rewareds		
leadsToPerformance	Badges, Levels, Feedback, Challenges, Theme, Leaderboards, Story, Rewards, Achievements	Performance	-